Nirapod Fine: An AI-Based Legal Advisory System for Traffic Violation Verification Using Official Traffic Regulations

Receipt Extraction, Deterministic Legal Verification, Manual-Review Routing and Integrity Protection

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Abstract

Traffic-monitoring systems can detect a violation, but a violation alone does not prove that the offence, the cited law and the fine amount are legally consistent. Nirapod Fine is an AI-assisted web system that verifies Bangladeshi traffic-fine receipts against the Road Transport Act, 2018. AI reads the receipt (image, PDF or URL); the user confirms the fields; and a deterministic rule engine checks the offence, section, fine limits, repeat offences and exceptions. Evaluation used 42 offence rules, 15 exception rules, 224 typed-field cases and 56 receipt images. The rule engine reached 71.4% verdict accuracy (macro F1 0.722); field extraction ranged from 66.1% to 96.4%; and the full image-to-verdict pipeline reached 46.4%. Uncertain cases were sent to manual review, with no false approvals in the image-based evaluation.

Introduction

Enforcement systems can issue notices, yet citizens cannot easily tell whether the offence, section and fine on a notice are lawful. Traffic AI mostly detects violations, while general legal AI may produce fluent but unsupported conclusions, and blockchain protects stored data without proving that a legal conclusion is correct. Nirapod Fine therefore separates three tasks: reading the document, reasoning about the law, and protecting the record.

OBJECTIVES

- Extract offence, fine, date, location, vehicle, receipt number and legal section from images, PDFs and URLs.
- Build a structured database of offences, fine limits and legal exceptions.
- Verify offence, section, fine amount, repeat offence and exceptions with a deterministic rule engine.
- Explain results from the validated output and send uncertain cases to human review.
- Protect datasets and results with hashing and optional blockchain storage.

RESEARCH QUESTIONS

- **RQ1:** How accurately can receipt information be extracted?
- **RQ2:** How effectively can a rule engine verify consistency with the 2018 Act?
- **RQ3:** How do extraction errors affect the final verdict?
- **RQ4:** How can uncertain cases be identified for human review?
- **RQ5:** How can results be protected without storing personal data on-chain?

Methodology

The system follows five steps. AI is used only to read the receipt and to word the explanation; the legal status always comes from the rule engine after the user has confirmed the fields.

1. Upload	The user submits an image, PDF or URL-based traffic receipt (Bangla, English or mixed).
2. Extract	AI reads offence, amount, date, location, vehicle, receipt number, section and notes.
3. Confirm	The user compares the fields with the receipt and corrects uncertain values.
4. Verify	The rule engine matches the offence, checks fine limits, repeat offences and exceptions.
5. Explain	A status and a constrained explanation are returned, or the case goes to manual review.

KEY FEATURES

- **Six statuses:** valid, fine too high, reason mismatch, exception possible, exception not applicable and manual review.
- **User-confirmed:** user edits override uncertain AI readings.
- **Safe by design:** AI never decides legal validity; explanations are checked against the rule output.
- **Versioned legal dataset** with hash-based audit.

Proposed Model

The system is built in layers: web front end, FastAPI back end, receipt processing, legal knowledge, rule engine, explanation service, and a database with an optional blockchain integrity layer.

Fig. 3.1. Overall architecture of the proposed traffic-fine verification system.

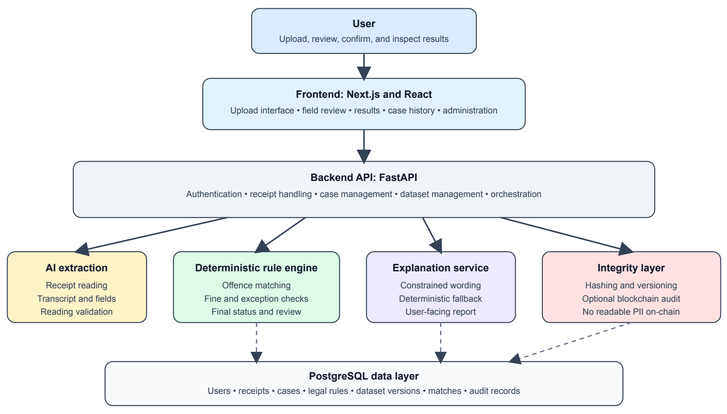


Figure 1: Overall architecture of the proposed system.

Data Analysis

42 offence rules	**15** exception rules	**2018** Road Transport Act

Rules were structured from the Road Transport Act, 2018 and a selected traffic-law exceptions guide. Each offence stores its section, description, fine limits, synonyms, keywords and conditions. Evaluation used 224 manually labelled typed-field cases (standard, boundary, paraphrased, Bangla, missing-amount and other groups) and 56 receipt images.

Four stages were evaluated: the typed-field rule engine, AI receipt reading, the full image-to-verdict pipeline, and an AI-only baseline for comparison.

Results

71.4% typed-field verdict accuracy	**0.722** macro F1-score
46.4% image-to-verdict accuracy	**7.7 s** mean time per receipt image

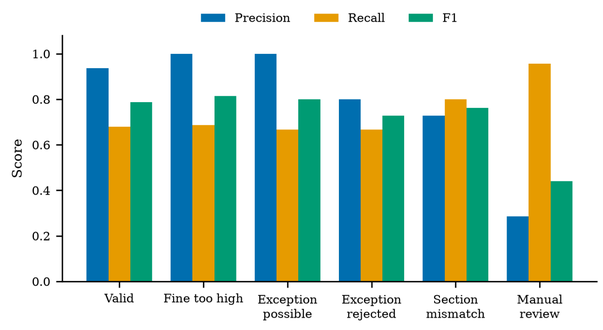


Figure 3: Verdict precision, recall and F1-score of the rule engine.

Accuracy was 100% for missing-amount, no-ceiling and below-minimum groups, but lower for paraphrased (59.3%) and Bangla (66.7%) descriptions. Top-three section matching reached 100% on standard cases.

System Tools and Technologies

Next.js, React, TypeScript front end with Tailwind CSS	**Python, FastAPI** REST API and business logic
PostgreSQL, SQLAlchemy data store with Alembic migrations	**Pillow, PyMuPDF** image preprocessing and PDF text
Clerk user and administrator access	**Solidity, Hardhat, Web3** SHA-256 hashing and AuditRegistry
External AI model receipt reading and explanations	**Docker, pytest, Vitest** services and testing

Discussion and Conclusion

Nirapod Fine shows a practical separation between AI-assisted receipt reading and deterministic legal verification. It gives explainable statuses, user confirmation and manual-review routing, and it is best described as a legal verification assistant, not an autonomous decision maker. Limitations: only the 2018 Act and a selected exceptions guide were covered; the evaluation set was small; the URL workflow was untested; and dates, amounts and Bangla text were error-prone. Future work will extend the legal dataset, collect varied real receipts, improve Bangla OCR, compare AI models and run user studies with drivers, officers and legal experts.

Results (continued)

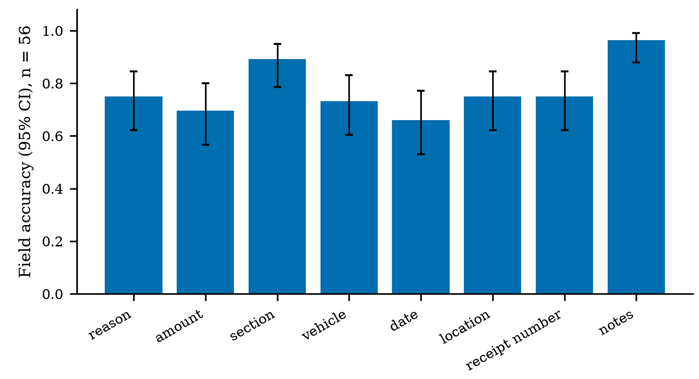


Figure 4: Field-level accuracy of AI receipt reading.

All 56 images were processed. Notes (96.4%) and section (89.3%) were read best; amount (69.6%) and date (66.1%) were weakest, which is why user confirmation is essential.

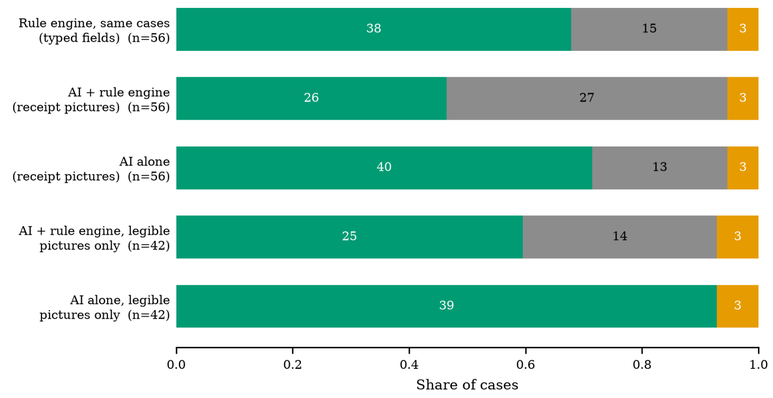


Figure 5: Decision quality of the rule engine, AI-only baseline and hybrid system.

End to end, 26 of 56 receipts were correct; 13 errors came from the rule engine, 12 from receipt reading and 5 from both. The AI-only baseline had higher direct accuracy (71.4% vs 46.4%), but it answers even when a receipt is incomplete. The hybrid system is more conservative: it made 26 correct decisions, referred 27 cases to manual review and made 3 confident errors, with 88.5% firm-decision accuracy and no false approvals.

Integrity and Explainable AI

Case hash confirmed receipt data and result	**Dataset hash** versioned legal rules	**Result hash** status and explanation

Only SHA-256 hashes and limited metadata are written to the optional blockchain audit contract; no readable receipt or personal data is stored on-chain. Hashes can be recomputed later to detect changes. AI only words the explanation, which is checked so that it cannot mention a section or amount that the rules did not produce.

The Tool in Action

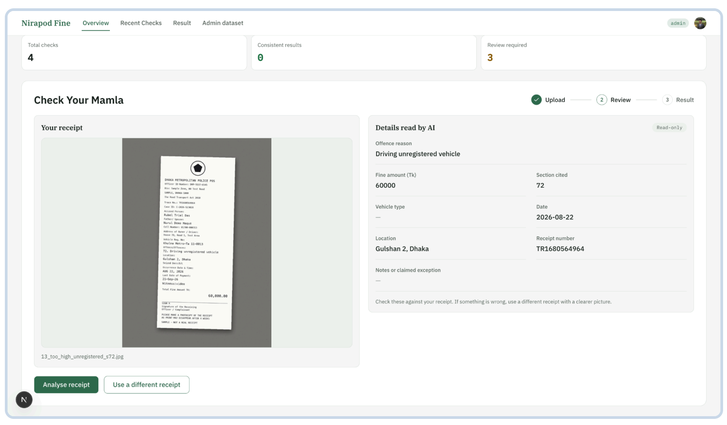


Figure 6: Review stage, where the user checks the details read by AI.

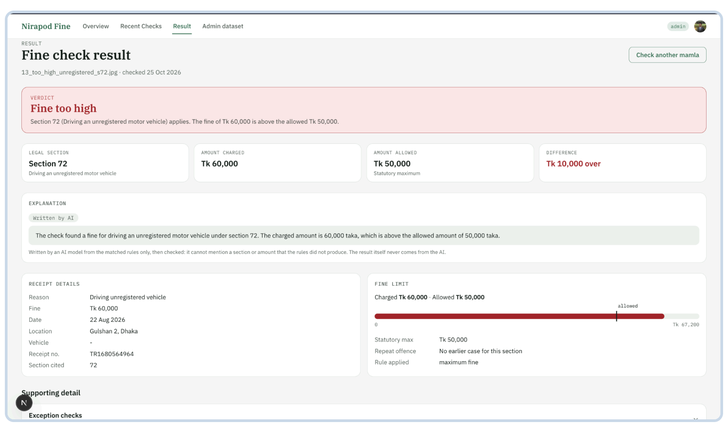


Figure 7: Result screen showing the verdict, legal section, allowed fine and explanation.

References

[1] World Health Organization, "Global status report on road safety 2023: Bangladesh," 2023.

[2] Asian Transport Observatory, "Bangladesh road safety profile 2025," 2025.

[23] Government of Bangladesh, Road Transport Act, 2018.

ACKNOWLEDGEMENT

We thank AIUB and the Department of CSE for their support.

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American International University-Bangladesh (AIUB)